

CAPSTONE PROJECT · ENTERPRISE ARCHITECTURE

SentinelAI

Enterprise AI-powered Security Operations Platform for automated threat ingestion, intelligent incident classification, risk prioritization, incident orchestration, and executive security reporting. Built using n8n, PostgreSQL, Gemini AI, and enterprise workflow automation.

7

LAYERS

5

WORKFLOWS

4

ACTORS

24/7

AUTOMATION



n8n



PostgreSQL



Gemini AI



Webhook API



HTML Dashboard



Gmail SMTP



Automation



AI Decision Engine

ARCHITECTURE HIGHLIGHTS

- ✓ 7 Enterprise layers
- ✓ 5 Automated workflows
- ✓ AI threat classification
- ✓ Real-time alert processing
- ✓ Automated incident escalation
- ✓ Executive dashboard reporting
- ✓ Enterprise audit trail

LAYER 1 Security event sources

Continuously generate raw security alerts



Firewall

Network perimeter traffic logs



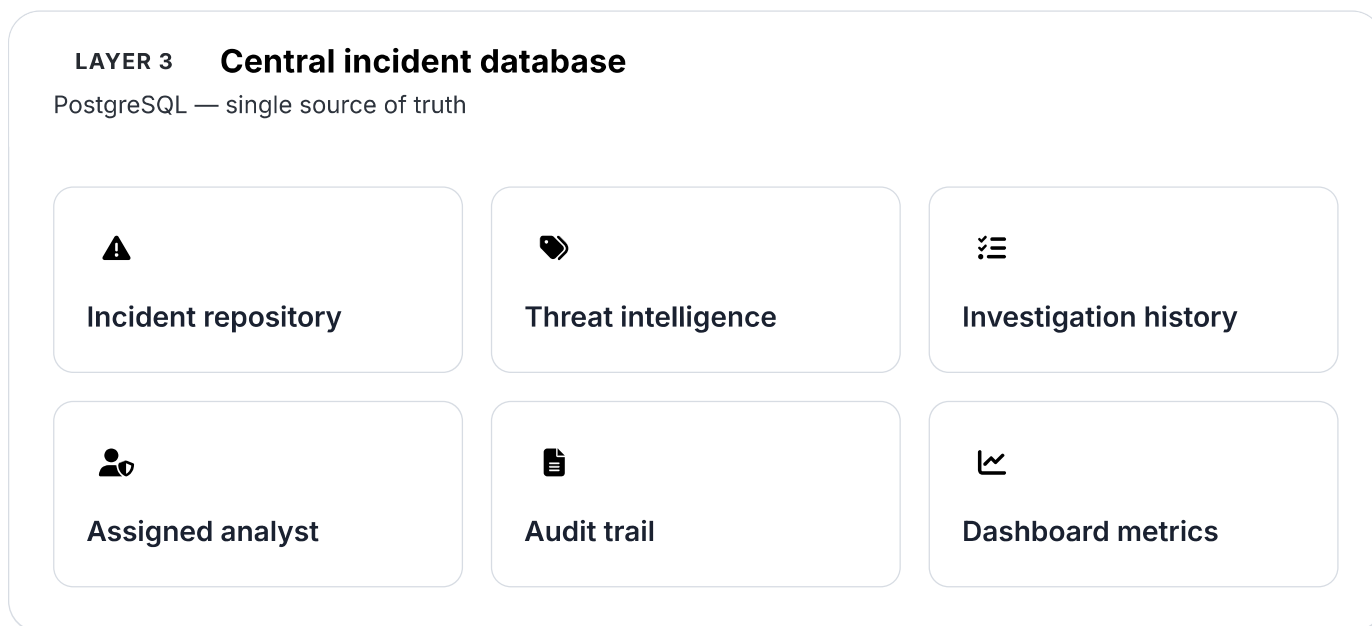
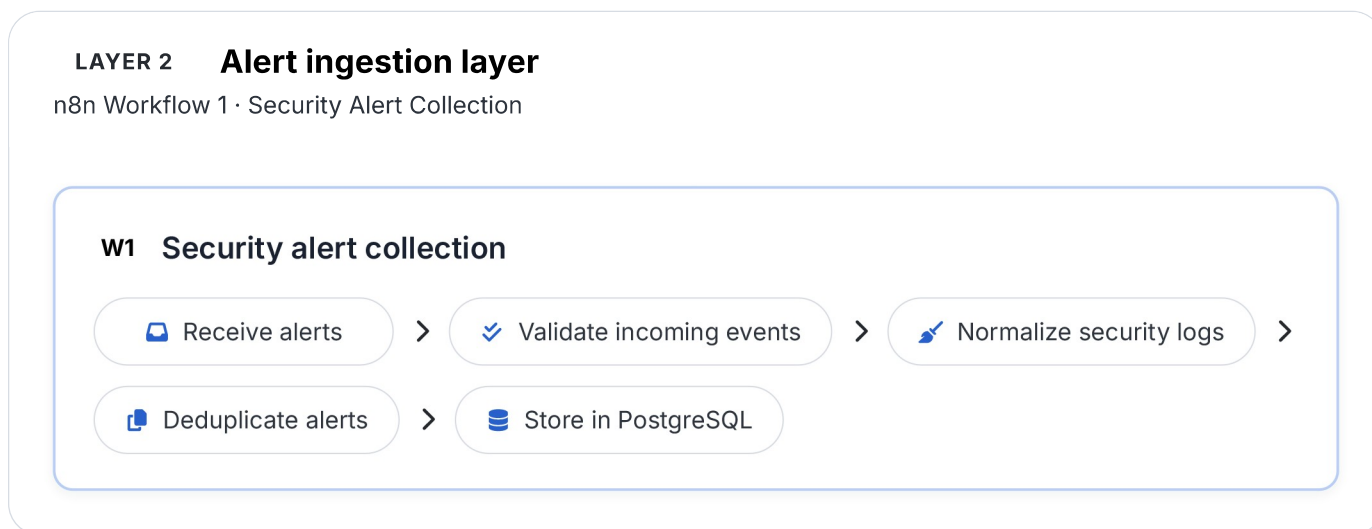
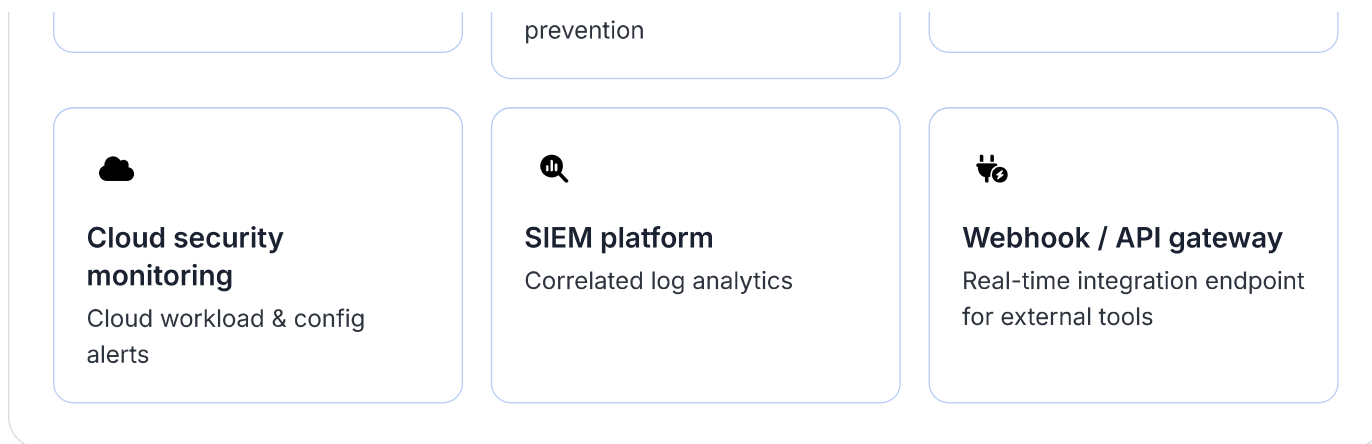
IDS / IPS

Intrusion detection &



Endpoint antivirus

Device-level threat signals



LAYER 4 AI intelligence layer

n8n Workflow 2 · AI Threat Classification & Prioritization

W2 AI threat classification & prioritization Fetch new alerts >  Send context to Gemini AI >  Update PostgreSQL

Threat category

Severity level

Priority score

Risk score

Recommended response

**LAYER 5 Incident response layer**

n8n Workflow 3 · Incident Assignment & Escalation

W3 Read classified incidents → decision engine**Critical****High****Medium****Low****CRITICAL INCIDENTS**

- > Immediate email notification
- > Escalation triggered
- > Assign security team
-  Human approval (security manager)

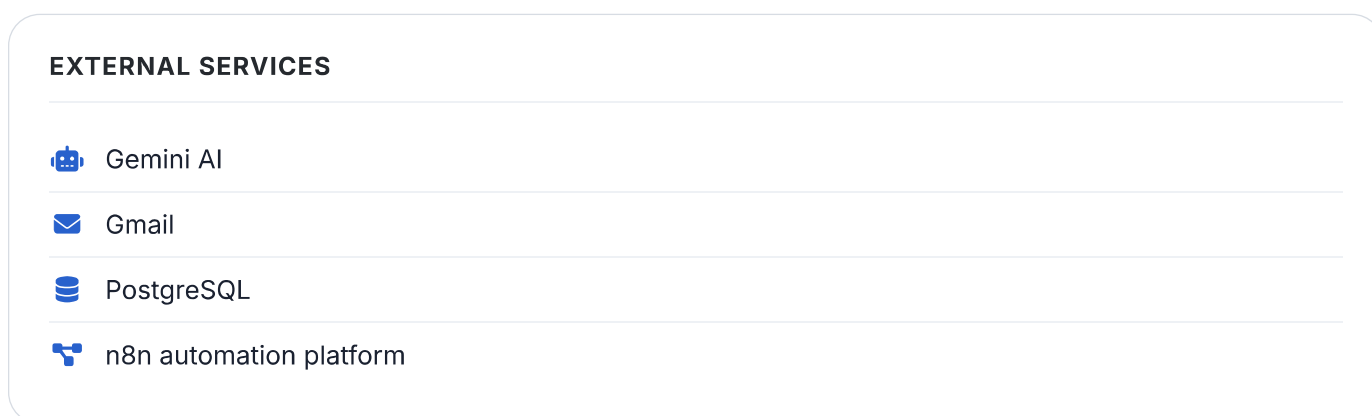
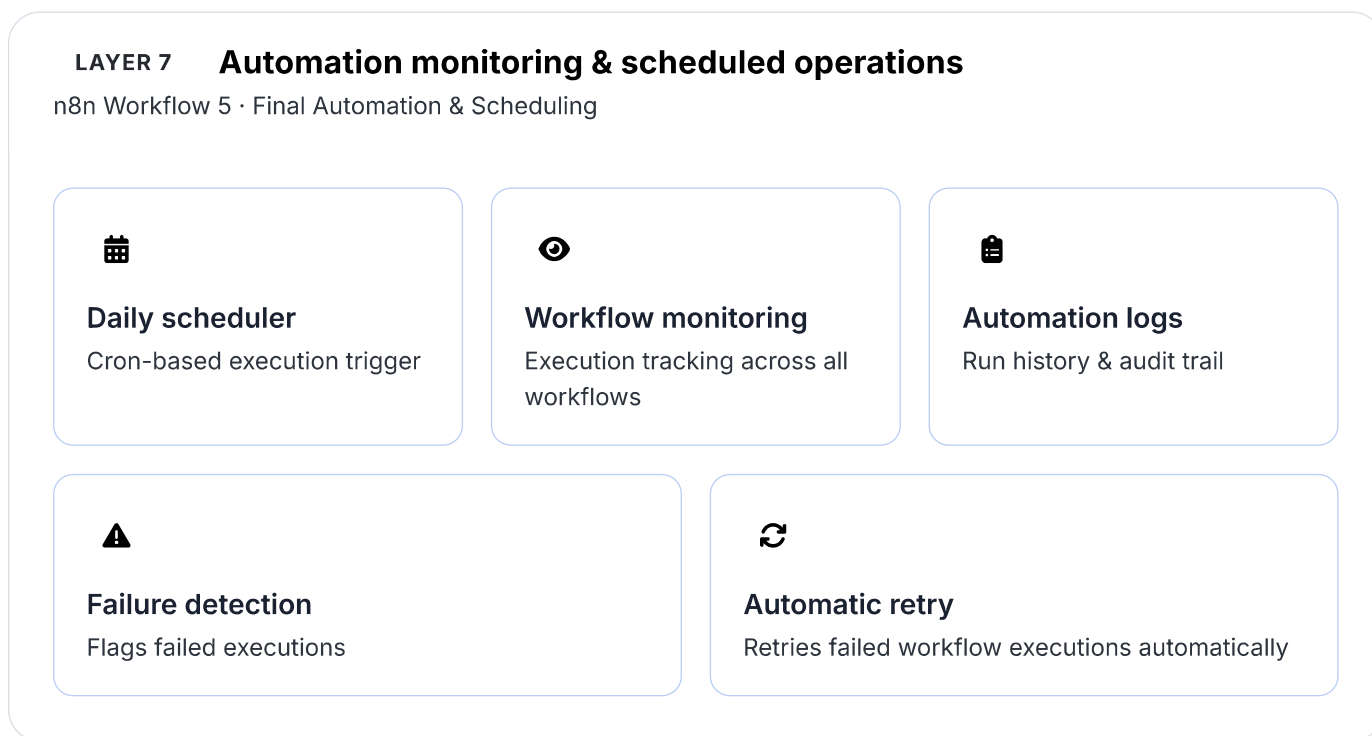
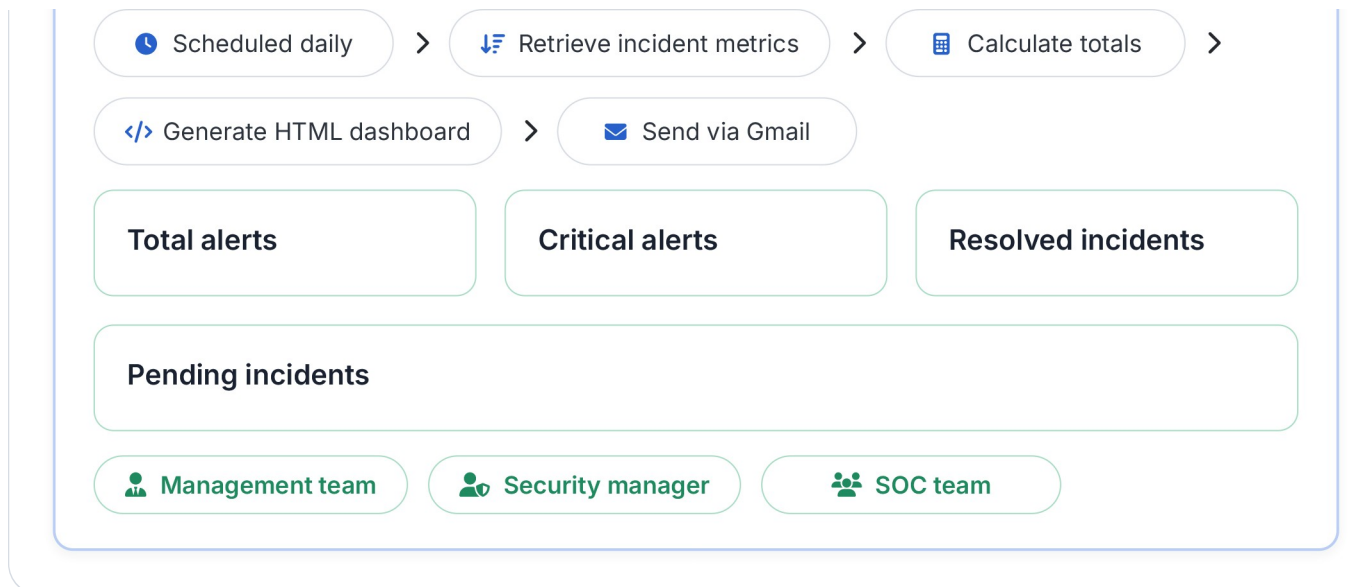
MEDIUM / LOW INCIDENTS

- > Assign security analyst
- > Update incident status

**LAYER 6 Security analytics & reporting**

n8n Workflow 4 · Executive Security Dashboard & Reporting (scheduled daily)

W4 Executive security dashboard & reporting



ACTORS

-  Security analyst
-  SOC team
-  Security manager
-  Organization management

ADDITIONAL COMPONENTS

-  Audit logging
-  Incident history
-  Email notifications
-  AI decision engine
-  Conditional routing
-  Database
-  Automation monitoring

Core workflow / source AI processing Critical path Reporting metric Data store / structural